PPP-E-1533B
March 14, 1975
SUPERSEDING
Fed. Spec. PPP-E-1533A
November 3, 1970

FEDERAL SPECIFICATION

ENVELOPES, PACKAGING, CUSHIONED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cushioned envelopes made from one piece of single face corrugated fiberboard or a draft paper/foam combination. The envelopes may be used for interior or exterior packaging when a light cushioning effect is required for items weighing not more than 10 pounds.

1.2 Classification.

1.2.1 Types. Envelopes covered by this specification shall be of the following types as specified.

Type I - Corrugated Fiberboard.

Style A - Manual.
Style B - Self Seal Enclosure.

Type II - Kraft Paper/Foam Combination - Self Sealing Closure.

1.2.2 Dimensions. The envelopes shall range in width from 4 inches to 14-3/4 inches and 6 inches to 20-1/2 inches in length (see table II).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-F-320	-	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade), and Cut Shapes.
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam.

FSC 8105

Source: https://assist.dla.mil -- Downloaded: 2026-04-26T12:51Z
Check the source to verify that this is the current version before use.

Federal Standards:

Fed. Test Method Std. No. 101 - Preservation, Packaging, and Packing Materials: Test Procedure.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Facing. The facing (i.e., outer liner) shall be made of the following materials.

3.1.1.1 Type I. Unbleached liner board having a minimum basis weight of 24 pounds per 1,000 square feet (NSF) for envelopes up to 12 inches in length by 9 inches in width and 33 pounds per NSF for envelopes of greater dimensions (see 4.4.1).

3.1.1.2 Type II. Seventy-four (74) pound kraft paper outer liner minimum 24 X 36 - 500 basis weight, see 4.4.1).

3.1.2 Cushioning medium.

3.1.2.1 Type I. The medium shall be made of unbleached corrugated paperboard weighing not less than 24 pounds per 1,000 square feet (see 4.4.1).

3.1.2.2 Type II. The medium shall be made of polypropylene foam conforming to PPP-C-1797 and shall be 3/32 inch or 1/8 inch nominal thickness as specified (see 6.2).

3.1.3 Adhesives.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:51Z
Check the source to verify that this is the current version before use.

3.1.3.1 Adhesives for combining the cushioning medium and liners and for sealing envelope seams shall conform to those commonly used for this purpose. At the manufacturer's option, the seams shall be heat sealed through the use of polyethelene coating applied to the cushioning medium.

3.1.3.2 For sealing type I, style B, and type II, a pressure sensitive adhesive shall be used.

3.2 Construction.

3.2.1 Type I.

3.2.1.1 Flute arrangement. The corrugating medium shall be either A, B, or C flute and the flutes shall be parallel to the envelope length as specified.

Note. A flute - 33 - 39 flutes per foot.
B flute - 47 - 53 flutes per foot.
C flute - 39 - 42 flutes per foot.

3.2.1.2 Combined board. The medium and facing shall be combined with a 100 percent fiber-tearing adhesive bend. The medium shall be on the inside of the envelope (see 4.3.1.1).

3.2.1.3 Envelope seams. The envelope shall be of either center seam or side seam construction. The seam width shall be not less than 11/16-inch nor more than 1-5/8 inches and the corrugations in the seam area shall be crushed to obtain a fiber-tearing adhesive bond over not less than 75 percent of the seam area (see 4.3.3). The facing shall not be broken along the inside seam edges or at any edge where it was folded to make a seam. Seams and flaps shall be formed to the rear, opposite the face, of the envelope.

3.2.1.4 Envelope flaps. Type I, style A envelopes shall be square cut and scored 1-1/2-inches ± 1/4-inch from the open end so that the end of the envelope may be folded over as a flap. Type I, style B envelopes shall have a flap extending 1-1/2 to 3-inches beyond the lip of the envelope and the flap shall be scored to fold completely over without breaking the facing. The closure shall be extended across the envelope width for both types.

3.2.1.5 Dimensions. Dimensions given shall be the inside dimensions with a tolerance of ± 1/4 inch (see 4.3.1.2).

3.2.1.6 Self-Sealing closure. The pressure-sensitive adhesive shall not stick to anything but itself and shall form a fiber-tearing bond over not less than 75 percent of the area of contact between the flap and body of the envelope (see 4.4.2).

3.2.2 Type II.

3.2.2.1 General requirements. The envelope shall be of the conventional open end, center seam style. The seams shall be folded and glued with water resistant adhesive. The envelope shall have an open end, tapered flap closure, effected by a pressure sensitive adhesive. The adhesive shall be protected with a carrier sheet, capable of being removed merely by stripping without exhibiting adhesive delamination. Adhesive shall be applied across the width of the tapered flap closure.

3.2.2.2 Additional requirements. Adhesive shall be placed between foam and draft paper in amounts and locations sufficient to secure foam. All paper to paper envelope overlap seams shall not be less than 5/8 inch nor more than 1-3/4 inches in width. Envelopes shall have a tapered flap closure extending a minimum of 1-1/2 inches beyond the throat of the envelope and extending the width of the envelope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:51Z
Check the source to verify that this is the current version before use.

3.2.2.3 Dimensions. Dimensions of the envelope shall be outside dimensions exclusive of the tapered flap closure. Tolerances shall be ± 1/4 inch.

3.3 Physical requirements.

3.3.1 Bursting strength (type I). The minimum average bursting strength shall be 60 pounds per square inch (p.s.i.) for single face board used to make envelopes up to 9-inches in width and 12-inches in length and 85 p.s.i. for single face board used to make envelopes of greater dimensions (see 4.4.3).

3.3.2 Seam and flap seal adhesion. The seams of all envelopes and flaps of type I, style B, and type II shall show not less than 75 percent fiber failure of the bonded area when tested as specified in 4.4.2.

3.4 Workmanship. Workmanship shall be first class throughout. Envelopes shall be uniformly constructed, free from holes, tears, cuts, splits or other defects which might impair their usefulness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services confrom to prescribed requirements.

4.1.1 Inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise specified.

4.2 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.1 Testing of components. The contractor shall certify in writing to the contracting officer or his representative that the liner and cushioning medium have been tested in accordance with 4.4.1, and meet the requirements of 3.1.1 and 3.1.2.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. An examination of the end item shall be made in accordance with classification of defects, inspection levels, and acceptable quality levels (AQL's) set forth below. A random sample shall be drawn from each lot offered for acceptance for visual and dimensional characteristics. For purpose of determining the sample size in accordance with MIL-STD-105 the lot size for examination under 4.3.2.1 through 4.3.1.3 shall be expressed in units of packages of envelopes of the same materials and for examinations under 4.3.1.4 shall be expressed in units of shipping containers.

4.3.1.1 Examination of the end item for defects in construction and workmanship. The sample unit for this examination shall be one envelope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:51Z
Check the source to verify that this is the current version before use.

Examine	Defects	
	Type I	Type II
Material	Not as specified (see 4.2.1)	Not as specified (See 4.2.1)
Appearance & Workmanship	Not squarely cut or glued. Tear, cut, scuff marks or dust. Seams not uniform.	Not squarely cut or glued. Tear, cut, scuff marks or dust. Seams not uniform.
Construction	Flaps not scored so as to effect a rold without distortion. Flutes not as specified (see 3.2.1).	No carrier sheet on pressure sensitive adhesive or flap closure. Foam thickness not as specified (see 3.1.2.2).
Adhesive	Corrugated medium not glued to facing as specified in 3.2.2 (peel apart to inspect).	Cushioning medium not glued to liner as specified in 3.2.2.2.

4.3.1.2 Examination of the end item for dimensional defects. The sample unit for this examination shall be one envelope.

Examine	Defects	
	Type I	Type II
Dimensions	Inside dimensions vary by more than ± 1/4 inch from those specified (see 3.2.5). Note. Inside dimensions may be checked using flat, rigid, rectangular templates made to the tolerances given in 3.2.5.	Outside dimensions vary by more than ± 1/4 inch from those specified in 3.2.2.3.
Seams	Seam dimensions not as specified (see 3.2.3).	Seam dimensions not as specified (see 3.2.2.2).
Flaps	Flap dimensions not as specified (see 3.2.4).	Flap dimensions not as specified (see 3.2.2.2).

4.3.1.3 Examination of preparation for delivery. An examination shall be made to determine that packing and marking comply with requirements of section 5. The sample unit shall be one shipping container fully prepared selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Examine	Defects
Packing	Not level specified, not in accordance with contract requirement. Shipping containers not fully prepared.
Marking	Marking illegible, incorrect, incomplete, omitted not in accordance with contract requirement.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:51Z
Check the source to verify that this is the current version before use.

4.3.1.4 Inspection levels acceptable quality levels (AQL's) for examination. The inspection levels, for determining the sample size, and the acceptable quality levels (AQL's), expressed in defects per 100 units, shall be S-2 with an AQL of 4.0 defects as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	S-2	4.0
4.3.1.2	S-2	4.0
4.3.1.3	S-2	4.0

4.3.2 Testing of end item. Testing shall be performed for the characteristics in table I on samples randomly selected from each lot of envelopes presented for examination. The sample unit shall be one envelope. Failure to meet the AQL, as indicated shall be cause for rejection of the lot.

TABLE I. End item testing

Characteristics	Requirements	Test Method	Applies to Sample lot Unit average	Results reported To nearest
Bursting strength (Type I)	3.3.1	4.4.3	X	p.s.i.
Seam adhesion (Type I, Type II)	3.3.2	4.4.2	X	percent
Flap adhesion (Type I, style B, Type II)	3.3.2	4.4.2	X	percent

4.4 Tests.

4.4.1 Basis weight test. In accordance with Fed. Test Method Std. 101, method 5022 (see 3.1.1 and 3.1.2).

4.4.2 Adhesion test. In accordance with Fed. Test Method No. 101, method 2002, except if no tensile testing machine is available the seams shall be separated by hand. The separated bonded area shall be examined for fiber failure expressed in percent of total bonded area. Specimens may be any representative size which can be cut from a seam or glued tapered flap closure.

4.4.3 Bursting strength test. In accordance with Fed. Test Method No. 101, method 2007, section 1.3. One thickness of material shall be tested from each sample unit, and no seam, fold or score line shall be included in the test specimen.

5. PREPARATION FOR DELIVERY

5.1 Packing. Envelopes shall be packed level A, B, or C, as specified in the procurement document.

5.1.1 Level A. Unless otherwise specified envelopes shall be packed in fiberboard boxes conforming to PPP-B-636, class weather-resistand, V3c or V3s.

5.1.2 Level B. Envelopes shall be packed in fiberboard boxes conforming to PPP-B-636, class

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:51Z
Check the source to verify that this is the current version before use.

domestic. The boxes shall be closed in accordance with PPP-B-636. Unless otherwise specified, the quantity, the quantity per shipping container shall be as specified in table II.

5.1.2.1 When specified (see 6.2), the shipping container shall be grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification. When specified the shipping container material shall also be grade V4s of PPP-F-320.

5.1.3 Level C. Envelopes shall be packed in shipping containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination.

TABLE II

TYPE I

Dimensions (inches) Length and Width	Quantity Per Carton Style A	Quantity Per Carton Style B
8 by 4	500	250
10 by 5-1/2	500	250
10 by 6-1/2	500	250
12 by 7-3/4	200	100
12 by 9	200	100
14-1/2 by 9	200	100
14-1/2 by 10	200	100
16-1/2 by 11	200	100
19-1/2 by 13	100	50
20-1/2 by 14-3/4	100	50

TYPE II

Dimensions (inches) Width & Length (Not including closure flap)	Quantity Per Carton
4 by 6-3/4	500
5 by 9-1/4	250
6 by 9	250
7-1/4 by 10-3/4	250
8-1/2 by 10-3/4	125
8-1/2 by 13-1/4	125
9-1/2 by 13-1/4	125
10-1/2 by 14-3/4	125

5.1.4 Standard pack (civil agencies). Packaging of cushioning envelopes as specified in paragraph 5.1.2 shall be considered the standard pack for civilian agency procurements.

5.2 Marking.

5.2.1 In addition to any special marking required by the procurement documents the shipping

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:51Z
Check the source to verify that this is the current version before use.

containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129 as applicable.

6. NOTES

6.1 Intended uses.

6.1.1 These envelopes are intended for use as mailing envelopes when used as exterior packaging. They are not designed to be weather-resistant. Whether used as exterior or interior packaging, the envelopes may be used when a light cushioning effect is needed, such as in the shipment of publications, tape cassettes, small parts, clothing, and small tools.

6.1.2 The type I, style A should be closed with gummed tape or staples.

6.1.3 The type I, style B, and type II envelopes are intended to save time and promote ease of handling on packing lines by eliminating the need for staples, gummed tape or heat seal equipment.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Types, sizes, styles, and quantities required.
(c) When wood, plywood or wirebound containers are required (see 5.1.1).
(d) When level B packing requires weather proof containers (see 5.1.2.1).
(e) Level of packing required.
(f) Type II - Foam thickness (see 3.1.2.2).

Military custodians:

Army - GL
Navy - SA
Air Force - 60

User activities:

Army - GI
Navy - AS, MS, SH, OS, YD

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing activity:

GSA - FSS

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:51Z
Check the source to verify that this is the current version before use.